



COMPETENCY STANDARD

FOR

MARINE ENGINE OPERATION AND MAINTENANCE

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Marine Engine Operation and Maintenance serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Shipbuilding sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Marine Engine Operation and Maintenance. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (60 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-MEO-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	20
SICIP-SBD-MEO-02-G	Perform Computations Using Basic Mathematical Concepts	1. Identify calculation requirements in the workplace 2. Select appropriate mathematical methods 3. Perform calculations	20
SICIP-SBD-MEO-03-G	Interpret Technical Drawings and Manuals	1. Select technical drawing 2. Interpret technical drawings. 3. Interpret operation & maintenance manuals	20
Total Hour			60 Hrs.

Sector Specific Competencies (50 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-MEO-01-S	Apply Green Practices in Shipbuilding sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-SBD-MEO-02-S	Use Hand and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean and maintain hand tools and power tools	20
SICIP-SBD-MEO-03-S	Carry Out Precision Checks and Measurements	1. Select the job to be checked and measured 2. Select measuring and checking tool/instrument 3. Perform measurements 4. Check and record measurement 5. Clean, maintain and store measuring instruments.	20
Total Hour			50 Hrs.

Occupation Specific Competencies (370 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-MEO-01-O	Demonstrate Knowledge of Marine Diesel Engines and Systems	1 Interpret working principles of two-stroke and four-stroke diesel engines 2 Illustrate major components and system of diesel engine 3 Explain operation of diesel engine	60
SICIP-SBD-MEO-02-O	Perform Combustion System Servicing	1 Carry out piston servicing and maintenance 2 Carry out cylinder head servicing and maintenance 3 Service cylinder liner 4 Perform main and crank pin bearing clearance check 5 Clean and store tools and equipment	90
SICIP-SBD-MEO-03-O	Perform Engine Oil System Servicing	1 Carry out engine oil replacement 2 Service engine oil pump 3 Service and maintain engine oil cooler	70
SICIP-SBD-MEO-04-O	Perform Cooling System Servicing	1 Carry out cooling chamber maintenance 2 Service cooling pump 3 Service and maintain water cooler	90
SICIP-SBD-MEO-05-O	Perform Maintenance of Intake Exhaust System	1 Carry out air cooler maintenance 2 Carry out exhaust manifold maintenance	60
Total Hours			0 Hrs.

COMPETENCY STANDARD: MARINE ENGINE OPERATION AND MAINTENANCE

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-MEO-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 <u>Emergency response plans and procedures</u> are responded to. 1.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual

	1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices (including PPE) 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COMPUTATIONS USING BASIC MATHEMATICAL CONCEPTS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-MEO-02-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, selecting appropriate mathematical methods, and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 <u>Calculation requirements</u> are identified based on the tasks and specifications. 1.2 Calculation requirements are understood. 1.3 <u>Workplace information</u> is interpreted to determine the necessary calculations. 1.4 Relevant workplace documents are identified to gather necessary data for calculations.
2. Select appropriate mathematical methods	2.1 <u>Appropriate calculation method</u> is identified based on the specific requirements of the task. 2.2 Units of measurement are considered when selecting the method for conversion. 2.3 Selected calculation method is applied to verify its effectiveness before proceeding with the final calculation.
3. Perform calculations	3.1 Appropriate <u>tools and instruments</u> are identified based on the requirements of the calculation. 3.2 Tools and instruments are checked for accuracy and calibration. 3.3 Computation is made using the identified tools and instruments. 3.4 Final result is checked to verify the correctness of the calculation before reporting.

Range of variables:

Variables	Range (may include but not limited to)
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: 1.3.1 Length 1.3.2 Width

	1.3.3 Height 1.3.4 Thickness 1.3.5 Diameter
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition 3.2 Subtraction 3.3 Division and Multiplication 3.4 Conversion of measurement 3.5 Percentage and ratio calculation 3.6 Geometric Formulas (Square, Rectangle, Circle, Sphere, etc.)
4. Tools and instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods, such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas 1.3 Mathematical language, symbols 1.4 Measuring units
2. Underpinning skill	2.1 Computing basic mathematics 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume, etc. using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified calculation requirements from workplace information. 1.2 selected appropriate method to carry out the calculation requirements. 1.3 completed calculations using appropriate tools/instruments.
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTERPRET TECHNICAL DRAWINGS AND MANUALS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-MEO-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate technical drawings and plans. It specifically includes the tasks of selecting technical drawing, interpreting operation and maintenance manuals.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select technical drawing	1.1 Appropriate <u>technical drawing</u> is selected based on the specifications and requirements of the job. 1.2 Selected drawing is checked to ensure that it accurately reflects the job requirements.
2. Interpret technical drawings.	2.1 Drawing components, assemblies are identified 2.2 Dimensions are identified according to job requirement 2.3 Clearances/tolerances are checked in accordance with workplace standard 2.4 <u>Instructions</u> are identified and followed accurately. 2.5 Material <u>specifications</u> are interpreted 2.6 Symbols in drawing are interpreted.
3. Interpret operation & maintenance manuals	3.1 Operation and maintenance manuals are collected and interpreted 3.2 Operation and maintenance manuals are followed.

Range of variables:

Variables	Range (may include but not limited to)
1. Technical drawing	1.1 Sketches 1.2 Manuals
2. Instructions	2.1 Note 2.2 Instruction 2.3 Special instruction 2.4 Precaution
3. Specifications	3.1 Product specifications 3.2 Method specifications 3.3 Material specifications

Curricular Content Guide

1. Underpinning knowledge	1.1 Technical drawing interpretation 1.2 Sequence of drawing 1.3 Methods of checking and applying drawing for work 1.4 Drawing selection and checking method to ensure
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	conformity to the job requirements. 1.5 Drawing components, assemblies 1.6 Identification of dimensions according to job requirement 1.7 Procedure of checking clearances/tolerances 1.8 Work instructions 1.9 Material specifications 1.10 Drawing symbols interpretation 1.11 Use of operation and maintenance manuals
2. Underpinning skill	2.1 Practicing workplace safety 2.2 Interpreting drawing, following operation and maintenance manuals, 2.3 Performing jobs in accordance with the drawing 2.4 Performing calculation as per drawing 2.5 Selecting and checking of drawing to ensure conformity to the job requirements. 2.6 Identifying drawing components and assemblies 2.7 Identifying dimensions according to job requirement 2.8 Checking clearances/tolerances in accordance with workplace standard 2.9 Following operation and maintenance manuals
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, Projector 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified dimension according to job requirement 1.2 maintained clearances and tolerances according to workplace requirement. 1.3 interpreted technical drawing 1.4 interpreted operation & maintenance manuals
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-MEO-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Basic concepts of green</u> are understood. 1.2 Energy efficiency principles are interpreted. 1.3 <u>Sustainable materials</u> are identified. 1.4 Pollution control methods are understood. 1.5 <u>Environmental certifications and standards</u> are identified. 1.6 <u>Green supply chains</u> are interpreted to minimize environmental impact.
2. Minimize resources use in the workplace	2.1 Resource consumption is identified in the Shipbuilding sector. 2.2 <u>Energy-efficient equipment</u> is identified and maintained. 2.3 Materials are sourced ensuring that sustainable and recyclable materials are prioritized. 2.4 Water usage is minimized through efficient management. 2.5 Production processes are optimized. 2.6 <u>Green technologies and innovations</u> are implemented in the workplace.
3. Implement waste management practices	3.1 <u>Waste types</u> identified in the Shipbuilding sector. 3.2 Waste segregation practices are interpreted. 3.3 Recycling programs are established through external recycling facilities. 3.4 <u>Waste reduction strategies</u> are understood. 3.5 Safe disposal practices are described. 3.6 <u>Waste-to-energy solutions</u> are explored to reduce landfill contributions.

Range of Variables

Variable	Range May include but not limited to:
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1. Basic concepts of green	1.1 Principles of Green Practices 1.2 Sources of Environmental Impacts 1.3 Resource Utilization 1.4 Sustainable Alternatives 1.5 Waste Management Practices 1.6 Green Habits at Work 1.7 Compliance and Regulations
2. Sustainable materials	2.1 Recycled Metals 2.2 Biodegradable Lubricants and Fluids 2.3 Composite Materials with Low Environmental Impact 2.4 Low-VOC Paints and Coatings 2.5 Eco-friendly Packaging Materials 2.6 Durable and Repairable Components
3. Environmental certifications and standards	3.1 ISO 14001: Environmental Management System (EMS) 3.2 LEED Certification (Leadership in Energy and Environmental Design) 3.3 RoHS (Restriction of Hazardous Substances) 3.4 REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) 3.5 Energy Star Certification 3.6 Eco-Labeling Standards 3.7 Green Building Codes & Local Environmental Compliance
4. Green supply chains	4.1 Environmentally Responsible Sourcing 4.2 Eco-Friendly Transportation and Logistics 4.3 Supplier Environmental Compliance 4.4 Sustainable Packaging and Distribution
5. Energy-efficient equipment	5.1 High-Efficiency Electric Motors (IE2/IE3/IE4 Rated) 5.2 Variable Frequency Drives (VFDs) 5.3 Energy-Efficient LED Lighting Systems 5.4 Energy-Star Rated Industrial Equipment 5.5 Heat Recovery and Insulation Systems
6. Green technologies and innovations	6.1 Renewable Energy Integration 6.2 Additive Manufacturing (3D Printing) 6.3 Smart Manufacturing Systems (Industry 4.0) 6.4 Waste-to-Energy Technologies 6.5 Low-Emission Welding and Cutting Equipment 6.6 Closed-Loop Cooling and Lubrication Systems 6.7 Eco-Innovation in Product Design
7. Waste types	7.1 Metal Waste 7.2 Plastic and Packaging Waste 7.3 Hazardous Waste 7.4 Electronic Waste (E-Waste) 7.5 Dust and Particulate Waste 7.6 Wood and Pallet Waste
8. Waste reduction strategies	8.1 5S Workplace Organization 8.2 Lean Manufacturing Practices

	8.3 Material Optimization and Nesting Techniques 8.4 Reusing and Repurposing Waste
9. Waste-to-energy solutions	9.1 Thermal Conversion (Incineration with Energy Recovery) 9.2 Gasification and Pyrolysis 9.3 Biomass Boilers Using Organic Waste 9.4 Use of Waste Oils as Alternative Fuel 9.5 Waste Heat Recovery Systems (WHRS) 9.6 On-Site Micro Energy Generation Units

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of pollution. 1.2 Pollution control technologies 1.3 Legal and environmental frameworks: overview of national environmental policies and local environmental compliance regulations. 1.4 Importance of ISO 14001 and other EMS (Environmental Management System) standards. 1.5 Classification of sustainable, recyclable, and hazardous materials used in production. 1.6 Strategies for minimizing water and energy use during material processing. 1.7 Waste hierarchy: reduce, reuse, recycle, recover, and disposal (5Rs). 1.8 Industrial waste-to-energy concepts.
2. Underpinning Skills	2.1 Measuring and analyzing energy and water consumption using meters. 2.2 Identifying and operating energy-efficient machinery. 2.3 Monitoring and logging resource consumption data. 2.4 Adjusting machine settings or workflow layouts. 2.5 Applying green inventory practices. 2.6 Segregating waste effectively into recyclable, hazardous, and general categories. 2.7 Implementing basic pollution control activities.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery

	4.4 Learning manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-MEO-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation is maintained using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Power tools are operated properly and safely to perform bench work.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter are removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to
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	manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT PRECISION CHECKS AND MEASUREMENTS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-MEO-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out precision checks and measurements. It specifically includes the tasks of selecting the job to be measured, selecting measuring and checking tools/instrument, performing measurements, checking & recording measurements and cleaning, maintaining and storing measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select the job to be checked and measured	1.1 Job is selected for measuring and checking 1.2 Required <u>dimensional measurement</u> is determined in accordance with drawing/plan 1.3 Required <u>physical conditions</u> are identified in accordance with drawing/plan 1.4 Required <u>geometrical dimensions</u> are identified in accordance with drawing/plan 1.5 Job drawing is used to select the measuring instruments.
2. Select measuring and checking tool/instrument	2.1 Appropriate measuring instruments is selected in accordance with job requirement. 2.2 <u>Direct and indirect measuring instruments</u> and <u>checking instrument</u> are identified 2.3 Applications of measuring device is determined. 2.4 Usability and accuracy of measuring device is checked and verified. 2.5 Measuring device is prepared for measurement. 2.6 Fits, Tolerance, clearance and limits are identified according to job requirements.
3. Perform measurements	3.1 Measurements are obtained using appropriate measuring instrument. 3.2 <u>Systems of measurements</u> are identified and converted where necessary. 3.3 Measurement is kept accurately in accordance to specification 3.4 Measurement is checked against job requirement 3.5 Physical conditions are checked in accordance with job requirements 3.6 Geometrical dimensions are checked in accordance with job specifications
4. Check and record	4.1 Measurements are checked and recorded in

measurement	accordance with workplace procedure 4.2 Measurement is interpreted, recorded and communicated to authority
5. Clean, maintain and store measuring instruments.	5.1 Dust and dirt are removed from the measuring instruments 5.2 Condition of measuring instruments are checked 5.3 Appropriate lubricant is applied after use and prior to storage 5.4 Measuring instruments are checked and calibrated 5.5 Measuring instruments are stored in accordance with workplace procedure.

Range of Variables

Variable	Range May include but not limited to:
1. Dimensional measurement	1.1 Length 1.2 Width 1.3 Depth 1.4 Diameter 1.5 Radius 1.6 Height
2. Physical condition	2.1 Roughness 2.2 Color 2.3 Smoothness 2.4 Surface finish 2.5 Flatness
3. Geometrical dimension	3.1 Angle 3.2 Circle 3.3 Triangle 3.4 Square 3.5 Rectangle 3.6 Hexagon shape 3.7 Pentagon shape
4. Direct measuring instruments	4.1 Set squares 4.2 Dial indicators 4.3 Steel tape 4.4 Steel rule 4.5 Meter rule 4.6 Calculator 4.7 Vernier slide caliper 4.8 Digital Vernier slide caliper 4.9 Micrometer (inch/millimeter) 4.10 Digital micrometer 4.11 Vernier bevel protractor 4.12 Spirit level 4.13 AVO meter(analogue/digital)

	4.14 Thermometers 4.15 Water meter 4.16 Gas meter 4.17 Simple protractor
5. Indirect measuring instrument	5.1 Outside caliper 5.2 Inside caliper 5.3 Bevel tri-square 5.4 Telescoping gage 5.5 Straight edge 5.6 Side bar 5.7 Trammel
6. Checking instrument.	6.1 Plug gauge 6.2 Snap gauge 6.3 Screw pitch gauge 6.4 Slip gauges 6.5 Feeler gauges 6.6 Screw pitch gauge 6.7 Slip gauge 6.8 Tri-square 6.9 Center gauge 6.10 Bevel tri-square
7. Systems of measurements	7.1 ISO standard 7.2 English system 7.3 Metric system

Curricular Content Guide

1. Underpinning Knowledge	1.1 Difference between measuring and checking 1.2 Types of measuring tools and their applications 1.3 Types of checking tools and their applications 1.4 Geometrical dimensions and tolerances 1.5 Method, procedure and techniques when taking linear Measurements 1.6 Methods, procedures and techniques when checking physical conditions of work pieces 1.7 Methods, procedures and techniques when checking geometrical dimensions of work pieces 1.8 Measurement conversion systems 1.9 Workplace record keeping procedures 1.10 Preventive maintenance for measuring and checking tools 1.11 Calibration and adjustment procedures for measuring and checking tools
2. Underpinning Skills	2.1 Determining required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 2.2 Measuring and checking linear and geometrical

	dimensions within the required tolerance in accordance to specification 2.3 Checking physical conditions using appropriate checking tool 2.4 Identifying and converting systems of measurements where necessary. 2.5 Recording measurements in accordance with workplace procedure 2.6 Interpreting and communicating measurement to authority 2.7 Applying appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 2.8 Checking condition of measuring instruments, calibrating and storing in accordance with workplace procedure
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 determined required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 1.2 measured and checked linear and geometrical dimensions within the required tolerance in accordance to specification 1.3 checked physical conditions using appropriate checking
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	tool 1.4 identified and converted systems of measurements where necessary. 1.5 recorded measurements in accordance with workplace procedure 1.6 interpreted and communicated measurement to authority 1.7 applied appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 1.8 checked condition of measuring instruments, calibrated and stored in accordance with workplace procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: DEMONSTRATE KNOWLEDGE OF MARINE DIESEL ENGINES AND SYSTEMS	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-MEO-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to demonstrate knowledge of marine diesel engines and systems. It specifically includes the tasks of Interpreting working principles of two-stroke and four-stroke diesel engines, illustrating major components and system of diesel engine and explaining operation of diesel engine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret working principles of two-stroke and four-stroke diesel engines	1.1 The functionality of two-stroke and four-stroke engines are described 1.2 Two-stroke and four-stroke cycle diesel engines are compared 1.3 Pressure-charging principles are explained 1.4 <u>Engine combustion characteristics</u> are illustrated 1.5 <u>Common combustion faults</u> are explained
2. Illustrate major components and system of diesel engine	2.1. <u>Major components of diesel engine</u> are identified 2.2. Functions of major components of diesel engine are explained 2.3. <u>Major engine systems</u> are identified and explained
3. Explain operation of diesel engine	3.1. Starting system of diesel engine is explained 3.2. <u>Operational parameters</u> of diesel engines are explained 3.3. Emergency isolation and stopping procedures are clarified

Range of Variables

Variable	Range (Includes but not limited to):
1. Engine combustion characteristics	1.1 Cylinder Pressure 1.2 Heat Release Rate 1.3 Ignition Delay and diesel knock 1.4 Combustion Duration 1.5 Maximum Pressure Rise Rate
2. Common combustion faults	1.1 Insufficient air and fuel supply 1.2 Incomplete combustion 1.3 Exhaust temperature rise

3. Major components of diesel engine	3.1 Cylinder Block, liner & Head 3.2 Piston Assembly (crown, connecting rod, ring) 3.3 Crankshaft and bearings 3.4 Camshaft
4. Major engine systems	4.1 Fuel System 4.2 Cooling System 4.3 Lubrication System 4.4 Electrical System 4.5 Air Intake & Exhaust System 4.6 Scavenging system
5. Operational parameters	5.1 Fuel temperature and pressure 5.2 Engine Oil temperature and pressure 5.3 Exhaust gas temperature 5.4 Scavenge air temperature 5.5 Cooling water temperature and pressure 5.6 Battery (Specific gravity of electrolyte, terminal voltage)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions and difference of two-stroke and four-stroke diesel engines 1.2 Principles of pressure charging 1.3 Characteristics of engine combustion 1.4 Major components and systems of diesel engine 1.5 Operational parameters of diesel engines
2. Underpinning Skills	2.1 Identifying Major components of diesel engine 2.2 Identifying Common combustion faults 2.3 Illustrating engine system 2.4 Demonstrating operational parameter
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulate) 4.2 Marine operations site 4.3 Marine diesel engines and systems 4.4 Tools, equipment and personal protective equipment 4.5 Appropriate workplace. 4.6 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted working principles of two-stroke and four-stroke diesel engines 1.2 illustrated major components and system of diesel engine 1.3 explained operation of diesel engine
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COMBUSTION SYSTEM SERVICING	Nominal Duration: 90 Hrs.	Unit Code: SICIP-SBD-MEO-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform combustion system servicing. It specifically includes the tasks of carrying out piston servicing and maintenance, carrying out cylinder head servicing and maintenance, servicing cylinder liner, performing main and crank pin bearing clearance check and cleaning and storing tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out piston servicing and maintenance	1.1 <u>Piston</u> of the engine is opened and disassembled 1.2 <u>Piston rings</u> are checked and serviced or replaced 1.3 Piston crown is cleaned as per job requirement. 1.4 Piston pin is opened and its clearance is checked and replaced 1.5 Piston is assembled and fitted back as per instruction manual
2. Carry out cylinder head servicing and maintenance	2.1 <u>Exhaust and inlet valves</u> are opened, cleaned, checked 2.2 Valves are lapped or replaced and fitted back to the cylinder head 2.3 Cooling space are cleaned
3. Service cylinder liner	3.1. <u>Cylinder liner</u> wear down is measured and checked 3.2. Cylinder liner is replaced if required 3.3. Mating face between cylinder liner and head is checked and lapped if required.
4. Perform main and crank pin bearing clearance check	4.1 Main and crank pin bearing are identified 4.2 Main and crank pin bearing clearance are checked as per instruction manual 4.3 Reasons for bearing wear down are investigated and rectified
5. Clean and store tools and equipment.	5.1 Workplace, tools and equipment are cleaned and maintained in accordance with workplace requirements 5.2 Waste materials are disposed in proper place. 5.3 Tools, equipment and finished products are stored safely in accordance with workplace procedures

Range of Variables

Variable	Range (Includes but not limited to):
1. Piston	1.1 Crosshead piston for two stroke engines 1.2 Trunk type piston for four stroke engines
2. Piston rings	2.1 Compression ring 2.2 Oil scraper ring
3. Exhaust and inlet valves	3.1 Stainless steel valve seat 3.2 Nimonic steel valve seat 3.3 Rocker arm and push rod
4. Cylinder liner	4.1 Uniflow cylinder liner 4.2 Cross flow liner

Curricular Content Guide

1. Underpinning Knowledge	1.1. Engine Components and Functions 1.2. Disassembly and Assembly Procedures of piston 1.3. Engine Valve System and Functions 1.4. Valve Lapping and Replacement 1.5. Cylinder Liner Function and Importance 1.6. Cylinder Liner Replacement Procedures 1.7. Mating Face Inspection and Lapping 1.8. Main and Crank Pin Bearings Function and Importance 1.9. Bearing Identification and Types 1.10. Investigation and Troubleshooting of Bearing Wear
2. Underpinning Skills	2.1 Performing disassembly and assembly the piston 2.2 Performing disassembly and assembly exhaust and inlet valves 2.3 Checking cylinder liner wear using appropriate tools 2.4 Inspecting and lapping the mating face between the cylinder liner and head 2.5 Identifying main and crank pin bearings in the engine 2.6 Checking bearing clearance using appropriate tools 2.7 Investigating causes of bearing wear
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 carried out piston servicing and maintenance 1.2 carried out cylinder head servicing and maintenance 1.3 serviced cylinder liner 1.4 performed main and crank pin bearing clearance check 1.5 cleaned and stored tools and equipment\
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ENGINE OIL SYSTEM SERVICING	Nominal Duration: 70 Hrs.	Unit Code: SICIP-SBD-MEO-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform engine oil system servicing. It specifically includes the tasks of carrying out engine oil replacement, servicing engine oil pump and Servicing and maintaining engine oil cooler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out engine oil replacement	1.1 Existing <u>engine oil</u> is drained and sump is cleaned 1.2 Engine oil filter is replaced and fresh oil is poured/renewed
2. Service engine oil pump	2.1 Engine oil main pump and priming pump are inspected 2.2 Engine oil pumps are opened and disassembled 2.3 Servicing of Engine oil pumps are performed and fitted back
3. Service and maintain engine oil cooler	3.1 <u>Engine oil cooler</u> is checked and opened 3.2 Engine oil cooler serviced and cleaned

Range of Variables

Variable	Range (Includes but not limited to):
1. Engine oil	1.1 Mineral-Based Two-Stroke Oil 1.2 Synthetic Two-Stroke Oil 1.3 Biodegradable Two-Stroke Oil 1.4 Mineral-Based Four-Stroke Oil 1.5 Synthetic Blend (Semi-Synthetic) Four-Stroke Oil 1.6 Fully Synthetic Four-Stroke Oil 1.7 High-Performance Diesel Marine Oil
2. Engine oil cooler	2.1 Shell and Tube Oil Cooler 2.2 Plate-Type Oil Cooler

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and properties of engine oils (mineral, synthetic, biodegradable, etc.). 1.2 Functions and components of engine oil systems, including pumps and coolers. 1.3 Proper procedures for oil replacement and system servicing. 1.4 Causes and effects of oil contamination and
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	degradation. 1.5 Workplace safety procedures and environmental regulations for oil disposal.
2. Underpinning Skills	2.1 Performing Draining and replacing engine oil efficiently. 2.2 Cleaning and servicing engine oil filters and sumps. 2.3 Performing Disassembling, inspecting, and servicing engine oil pumps. 2.4 Checking, cleaning, and maintaining engine oil coolers. 2.5 Using appropriate tools and following servicing procedures accurately.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 carried out engine oil replacement 1.2 serviced engine oil pump 1.3 serviced and maintained engine oil cooler
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COOLING SYSTEM SERVICING	Nominal Duration: 90 Hrs.	Unit Code: SICIP-SBD-MEO-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform cooling system servicing. It specifically includes the tasks of carrying out cooling chamber maintenance, servicing cooling pump and servicing and maintaining water cooler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out cooling chamber maintenance	1.1 <u>Cooling chamber</u> is inspected 1.2 Cooling chamber is cleaned and descaled 1.3 Sealing rings are checked and replaced if required
2. Service cooling pump	2.1 <u>Cooling pump</u> is checked and opened 2.2 Cooling pump is serviced 2.3 Cooling pump is fitted back
3. Service and maintain water cooler	3.1 <u>Water cooler</u> is checked and opened 3.2 Water cooler is serviced as per the standard operating procedure 3.3 Water cooler is cleaned

Range of Variables

Variable	Range (Includes but not limited to):
1. Cooling chamber	1.1 Cylinder Cooling Chamber 1.2 Piston Cooling Chamber 1.3 Exhaust Valve Cooling Chamber 1.4 Turbocharger Cooling Chamber
2. Cooling pump	2.1 Centrifugal Pump 2.2 Gear Pump 2.3 Reciprocating Pump 2.4 Jet Pump (Eductor Pump)
3. Water cooler	3.1 Plate type 3.2 Shell and tube type

Curricular Content Guide

1. Underpinning Knowledge	1.1 Cooling System Components and Functions 1.2 Cooling Chamber Inspection and Maintenance 1.3 Cooling Pump Maintenance 1.4 Water Cooler Servicing and Cleaning 1.5 Standard operating procedures for cooling system
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	servicing. 1.6 Workplace safety and environmental regulations related to cooling systems.
2. Underpinning Skills	2.1 Inspecting, cleaning, and descaling cooling chambers. 2.2 Checking, servicing, and reassembling cooling pumps. 2.3 Inspecting, cleaning, and maintaining water coolers. 2.4 Using appropriate tools and adhering to servicing procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 carried out cooling chamber maintenance 1.2 serviced cooling pump 1.3 serviced and maintained water cooler
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MAINTENANCE OF INTAKE EXHAUST SYSTEM	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-MEO-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform maintenance of intake exhaust system. It specifically includes the tasks of carrying out air cooler maintenance and carrying out exhaust manifold maintenance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out air cooler maintenance	1.1 Air cooler is checked and opened 1.2 Air cooler is serviced as per SOP 1.3 Air cooler is cleaned and fitted back
2. Carry out exhaust manifold maintenance	2.1 <u>Exhaust manifold</u> is checked and cleaned 2.2 Exhaust manifold expansion bellows is checked and 2.3 Exhaust manifold expansion bellows serviced as per SOP

Range of Variables

Variable	Range (Includes but not limited to):
1. Exhaust manifold	1.1 Water-Jacketed Exhaust Manifold 1.2 Dry-Type Expansion Chamber Manifold

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions and types of air coolers and exhaust manifolds. 1.2 Proper cleaning and servicing techniques for intake and exhaust systems. 1.3 Common issues and failures in air coolers and exhaust manifolds. 1.4 Manufacturer guidelines and standard operating procedures (SOPs). 1.5 Workplace safety standards for handling exhaust system components.
2. Underpinning Skills	2.1 Inspecting, cleaning, and maintaining air coolers. 2.2 Checking and servicing exhaust manifolds and expansion bellows. 2.3 Following standard operating procedures (SOP) for maintenance tasks. 2.4 Reassembling components correctly after servicing.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 carried out air cooler maintenance 1.2 carried out exhaust manifold maintenance
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Marine Engine Operation and Maintenance
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer with internet connection	1
2.	Projector and projector screen	1
3.	Marine Diesel engine	3
4.	Engine Cooler (Tube type)	1
5.	Engine Cooler (Plate type)	1
6.	Engine Crank shaft	1
7.	Different engine component set	2
8.	Centrifugal Pump	2
9.	Positive displacement pump	2
10.	Main bearing	1
11.	Crank pin bearing	1
12.	Battery (100 AH) with charger	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Marine Engine Operation and Maintenance the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.